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Methodology and Exploratory Analysis of Stellar Host Parameters and Exoplanet Systems via Principal Component Analysis

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1. RESEARCH QUESTION

The primary research question guiding this analysis is: *How do stellar host parameters influence exoplanet system properties?* This question seeks to understand the relationships between the intrinsic properties of stars that host exoplanets and the characteristics of the planetary systems they harbor. By analyzing a comprehensive dataset of stellar parameters and system parameters, we aim to identify patterns and correlations that can shed light on the processes of planet formation and evolution, as well as the potential habitability of exoplanets based on their host star's characteristics.

As a preliminary step towards answering this question, an exploratory analysis using principal component analysis (PCA) was conducted on the stellar host parameters. The PCA uncovered known astrophysical phenomena, such as the main-sequence structure of the host stars and the age-metallicity relation, as well as an inverse relationship between planet count and star count in a system. These findings suggest that there are meaningful patterns in the data that can be further explored to understand the influence of stellar host parameters on exoplanet system properties.

Further analyses will include performing PCA on the stellar parameters alone to identify the main components of variance, and then projecting the system parameters onto these components to analyze their relationships. Additionally, data imputation techniques will be applied to address the issue of missing values in the dataset, allowing for a more comprehensive analysis that accounts for observational biases and the non-random nature of the missing data.

2. DATASET AND PREPROCESSING

The data set used for the principal component analysis (PCA) was obtained from the NASA Exoplanet Archive *STELLARHOSTS* table (NASA Exoplanet Archive 2026)[1]. The set of parameters extracted for this analysis is summarized in Table 1. The *STELLARHOSTS* table provides stellar parameters for all systems hosting confirmed exoplanets. For systems comprising multiple gravitationally bound stars, the parameters for each star are included. System parameters are denoted with the `sy_` prefix, while stellar parameters are denoted with the `st_` prefix.

Because the table includes a separate row for each reference, a host star with multiple published references can appear in multiple rows. To prepare the data for PCA, a data-cleaning procedure was applied to retain only one representative row per host star. The selection procedure was as follows:

1. The publication year of each stellar parameter entry was extracted from the `st_refname` field, and a new column, `st_ref_year`, was added to the data set to store this information.

[1] Accessed on 2026-02-22 at 11:35:01, returning 47,048 rows.

Parameter	Description
sy_name	System Name
hostname	Host Name
sy_snum	Number of Stars
sy_pnum	Number of Planets
st_refname	Stellar Parameter Reference
st_spectype	Spectral Type
st_teff	Stellar Effective Temperature [K]
st_rad	Stellar Radius [Solar Radius]
st_mass	Stellar Mass [Solar mass]
st_met	Stellar Metallicity [dex]
st_metratio	Stellar Metallicity Ratio
st_lum	Stellar Luminosity [log(Solar)]
st_logg	Stellar Surface Gravity [log10(cm/s^2)]
st_age	Stellar Age [Gyr]
st_dens	Stellar Density [g/cm^3]
st_vsin	Stellar Rotational Velocity [km/s]
st_rotp	Stellar Rotational Period [days]
st_radv	Systemic Radial Velocity [km/s]

Table 1. Data parameters collected from the NASA Exoplanet Archive STELLARHOSTS Table. More information can be found at https://exoplanetarchive.ipac.caltech.edu/docs/API_STELLARHOSTS_columns.html.

2. Parameter completeness was computed for each row, defined as the count of non-null values among the following columns: `st_teff`, `st_rad`, `st_mass`, `st_met`, `st_lum`, `st_logg`, `st_age`, `st_dens`, `st_vsin`, `st_rotp`, and `st_radv`.

3. For each star, the row corresponding to the most recent publication year was retained. In cases of identical or missing publication years, the row with the highest parameter completeness was selected. Remaining ties were resolved by random selection.

4. The ratio used in each row of `st_met` was defined in `st_metratio`. Based on this, `st_met` was also split into four separate columns: `st_met_FeH`, which contains the [Fe/H] metallicity value, `st_met_MH`, which contains the [M/H] metallicity value, `st_met_NH`, which contains the [N/H] metallicity value, and `st_met_mH`, which contains the [m/H] metallicity value. The `st_met` column was then dropped from the data set.

After applying this filtering procedure, the data set was reduced to 5,008 rows, with each row representing a unique host star and its most recent, most complete set of stellar parameters. The resulting cleaned data set was then used for PCA on the selected stellar parameters.

3. PRINCIPAL COMPONENT ANALYSIS

Principal component analysis (PCA) is a dimensionality reduction technique that transforms a set of correlated variables into a smaller number of uncorrelated variables called principal components.

Parameter	# NaN	% NaN
sy_pnum	0	0.00%
sy_snum	0	0.00%
st_teff	999	19.95%
st_rad	1040	20.77%
st_mass	1337	26.70%
st_met_FeH	2395	47.82%
st_logg	2793	55.77%
st_age	2951	58.93%
st_dens	3338	66.65%
st_lum	3942	78.71%
st_vsin	4088	81.63%
st_radv	4590	91.65%
st_rotp	4619	92.23%
st_met_MH	4954	98.92%
st_met_mH	4996	99.76%
st_met_NH	5003	99.90%

Table 2. Distribution of missing values (NaN counts and percentages) across numerical parameters in the cleaned dataset. Parameters are sorted by increasing proportion of missing values.

For the purposes of this project, PCA was applied as an exploratory tool to identify underlying patterns and relationships among the stellar parameters in the dataset. By analyzing the loadings of each principal component, insights can be gained into which parameters contribute most strongly to the variance in the data, potentially revealing clusters of similar host stars or correlations between certain stellar properties.

To perform PCA, a dataset of n items with p features must be represented as a matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$. The columns of $\mathbf{X}$ are then standardized and the covariance matrix $\mathbf{C}$ of the data can be computed:

$$\mathbf{C} = \frac{1}{n-1}\mathbf{X}^T\mathbf{X} \tag{1}$$

The eigenvalues and eigenvectors of $\mathbf{C}$ are then calculated. The eigenvectors define the directions of maximum variance in the data (i.e., the principal component axes), while the corresponding eigenvalues quantify the variance explained by each component. Analyzing the loadings of each component reveals which original features contribute most strongly to the captured variance, thereby offering insight into the underlying structure of the data.

From the *STELLARHOSTS* table, categorical parameters `sy_name`, `hostname`, `st_refname`, `st_spectype`, and `st_metratio` were excluded from the PCA. The issue arose in choosing which of the remaining numerical parameters to include in the analysis, as some parameters had a high proportion of missing values. Table 2 shows the distribution of missing values across the numerical parameters, sorted by increasing proportion of NaN values. A number of solutions were considered to address the missing data.

3.1. *Complete Case PCA*

Given the high proportion of missing values in many of the numerical parameters, a complete case PCA approach was adopted to perform an initial, exploratory analysis. This method involved selecting a subset of the data that contained complete cases for a chosen set of parameters, allowing a trial run of PCA for initial results to be performed without the need for imputation. A NaN limit τ was defined, and only parameters with a proportion of missing values less than τ were included in the PCA. The raw results of all complete case PCA runs are included in Appendix A.

Complete case PCA with $\tau = 0.50$ yielded 2,359 rows of data with 5 principal components that together explained 93.21% of the variance.

1. **Principal Component 1 (24.70% variance explained)** was strongly influenced by mass (0.72) and radius (0.53), and mildly influenced by temperature (0.34) and the planet count(-0.25). This is a clear main-sequence star component, ranging from large, hot, massive stars to smaller, cooler stars. Planet count has in inverse relationship with the size, mass, and temperature. This could be due to observational biases, as larger, hotter stars tend to be more luminous making transit detections and direct imaging difficult. More massive stars also are less influenced by their exoplanets' gravity, making radial velocity detections difficult as well.

2. **Principal Component 2 (21.54% variance explained)** was strongly influenced by temperature (-0.61), metallicity (-0.60), and radius (0.51). This component may show an inverse relationship between stellar host size and temperature/metallicity. This is a strange result, as larger stars are typically hotter and more metal-rich. This can be also attributed to observational bias. Detecting planets around larger, hotter stars is more difficult, so the sample of host stars with measured parameters would not include many of these.

3. **Principal Component 3 (17.02% variance explained)** was strongly influenced by the number of stars (0.81) and the number of planets (-0.52). This component may represent an inverse relationship between systems with more gravitationally bound stars and systems with fewer planets, likely due to the gravitational interactions in multi-star systems that can disrupt planet formation and stability.

4. **Principal Component 4 (15.83% variance explained)** was strongly influenced by the number of planets (-0.81) and the number of stars (-0.55). This component captures the difference between systems with few companion stars and planets and systems with more of both.

5. **Principal Component 5 (14.13% variance explained)** was strongly influenced by metallicity (0.79) and temperature (-0.54). This component clearly represents an inverse relationship between metallicity and temperature. This is a known relationship in stellar astrophysics, recently explored by Martínez-Hernández et al. (2026).

Overall, PC1 and 2 are likely due to observational biases against larger, hotter stars. PC3 and 4 show the variance in exoplanet systems themselves, with PC3 showing the influence of gravitational interactions in multi-star systems and PC4 showing the influence of stellar mass on planet formation. PC5 shows a known relationship between metallicity and temperature. These results suggest that the PCA is capturing meaningful patterns in the data, but also highlights the limitations of the dataset due to missing values and observational biases.

Complete case PCA with $\tau = 0.70$ yielded 583 rows of data with 6 principal components that together explained 95.88% of the variance. The results were similar to the $\tau = 0.50$ case, but with some differences in the loadings and variance explained by each component.

1. **Principal Component 1 (44.53% variance explained)** was strongly influenced by mass (0.48), radius (0.42), temperature (0.44), log g (-0.46), and mildly influenced by density (-0.37). This is a clear main-sequence star component, ranging from smaller, cooler stars with higher surface gravity and density to larger, hotter stars with lower surface gravity and density. The system parameters contribute weakly, suggesting that most variation in the data is due to intrinsic stellar properties. This component is similar to PC1 in the $\tau = 0.50$ case, but with no clear correlation with the number of planets.

2. **Principal Component 2 (12.80% variance explained)** was strongly influenced by age (-0.59), metallicity (-0.53), number of stars (0.49). This component ranges from young, metal-poor stars with more stellar companions to older, metallic stars with fewer companions. This is another astrophysical component, with only weak interactions with the system parameters. This may also be an orthogonal component to the age-metallicity relation, which is a known phenomenon explored by Twarog (1980) and Edvardsson et al. (1993).

3. **Principal Component 3 (12.19% variance explained)** was strongly influenced by the number of planets (-0.74) and the number of stars (0.56). This is a similar component to PC3 in the $\tau = 0.50$ case, showing the influence of gravitational interactions in multi-star systems on planet formation and stability.

4. **Principal Component 4 (10.44% variance explained)** was strongly influenced by age (0.67) metallicity (-0.52), and mildly influenced by the number of stars (0.40). This component clearly shows an inverse relationship between age and metallicity and is orthogonal to PC2. This is likely the AMR.

5. **Principal Component 5 (8.70% variance explained)** was strongly influenced by metallicity (-0.53) number of planets (-0.57), and number of stars (-0.52). This component is similar to PC4 in the $\tau = 0.50$ case, simply differentiating between sparse and populated star systems, but also includes shows a strong influence of metallicity proportional to the number of stellar companions and exoplanets.

6. **Principal Component 6 (7.12% variance explained)** was strongly influenced by density (0.60) and radius (0.57), and mildly influenced by log g (-0.31) and temperature (-0.33). This is a weak component that is orthogonal to the main-sequence component (PC1). This is probably present due to differences in stellar structure or evolutionary stage that are not captured by the other parameters.

With more components falling within the threshold, the number sample size decreased significantly, but the results were more astrophysically meaningful. PC1 and PC6 together capture the main-sequence structure of the host stars, while PC2 and PC4 capture the age-metallicity relation. PC3 and PC5 show the influence of stellar companions on planet formation, with PC5 also showing a strong influence of metallicity.

Raising τ higher than 0.70 greatly diminished the sample size with respect to data dimensionality[2], and the results were not as meaningful.

Though it contained more samples, $\tau = 0.50$ had more components that were likely due to observational biases, while $\tau = 0.70$ had fewer samples but more astrophysically meaningful components. This highlights the trade-off between sample size and data quality when performing PCA on datasets with missing values. The complete case PCA approach provided valuable insights into the underlying structure of the data, but also underscored the limitations of the dataset and the need for careful interpretation of the results.

Both complete case PCA runs uncovered known astrophysical phenomena and also suggested an inverse relationship between planet count and star count in a system. Though the majority of variance is from intrinsic stellar properties, the system properties also show a non-negligible influence on the variance in the data, suggesting that the dataset contains meaningful patterns related to stellar properties and their influence on exoplanet systems.

3.2. *Extensions to PCA*

This preliminary analysis demonstrated the likelihood of finding underlying patterns in the stellar host data, but did not directly answer the research question of how stellar host parameters influence exoplanet system properties. To address this, the system parameters can be dropped and complete case PCA can be performed again to identify the main components of variance in the stellar parameters alone. Then, the system parameters can be projected onto these components to analyze how they relate to the underlying structure of the stellar host properties. This approach would allow for a more direct analysis of the influence of stellar host parameters on exoplanet system properties, while still accounting for the limitations of the dataset.

A selected number of stars with no exoplanets can also be included in the analysis to provide a more comprehensive view of the relationships between stellar properties and planet formation. This would allow for a comparison between stars with and without exoplanets, potentially revealing differences in stellar properties that are conducive to planet formation. The number of non-host stars included should be carefully chosen to balance the representation of different stellar types and to avoid introducing additional biases into the analysis.

However, the high proportion of missing values and observational biases limit the conclusions that can be drawn from complete case PCA. To address this issue, the missing data must be imputed and the full dataset must be artificially extended to perform more comprehensive PCA on a larger number of parameters. This must be done with observational biases in mind, as the missing data is not random, and the possibility of undetected exoplanets exists for all star systems.

4. DATA IMPUTATION THROUGH PSEUDO-GIBBS SAMPLING

Missing stellar host data can be attributed to a variety of factors, including observational limitations, selection biases, and the inherent difficulty of measuring certain stellar parameters. For all missing data, true values exist, and their missingness is likely due to a mixture of their own or other parameters' values, as well as technological difficulty.

GSimp, a Gibbs sampler-based imputation method developed by Wei et al. (2018), is a powerful tool for imputing missing not at random (MNAR) data in datasets with complex relationships between

[2] Assuming a well-sampled dataset has 2^D samples where D is the number of dimensions.

variables. A modified version of GSimp can be applied to the stellar host dataset to impute missing values while accounting for the underlying structure of the data and the non-random nature of the missingness.

GSimp acts upon a data matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ with missing values. Wei et al. (2018) initializes missing values using Quantile Regression Imputation of Left-Censored Data (QRILC), a method designed specifically for left-censored data (i.e., values known to fall below a detection limit). However, in the stellar host dataset, missing values often arise from observational biases and limitations in addition to true censoring. Therefore, the stellar host dataset may not be generalized as purely left-censored, and QRILC may not be the most appropriate initialization method for all missing values.

Instead, we initialize missing values by drawing from the known parameter distributions across all stellar types. This approach offers two key advantages: (1) it accounts for observational biases that underrepresent certain stars in the dataset, and (2) it allows the analysis to consider exoplanets around underrepresented stellar types under the assumption that planets may form around all classes of stars.

The notation for describing the algorithm is defined as follows:

- In data matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, the values in the current feature p are denoted as column vector y_p of length n. The values in the other features are denoted as matrix $\mathbf{y}_{-p} \in \mathbb{R}^{n \times (p-1)}$.
- The indices of missing values in y_p are written as mp, and the indices of observed values are written as op. Therefore, the column vector of missing values in feature p is denoted as $y_{mp,p}$ with length $n_{miss,p}$, and the column vector of observed values in p is denoted as $y_{op,p}$ with length $n_{obs,p}$.
- Values of other features $-p$ corresponding to $y_{mp,p}$ are denoted as matrix $\mathbf{y}_{mp,-p} \in \mathbb{R}^{n_{miss,p} \times (p-1)}$, and values of $-p$ corresponding to $y_{op,p}$ are denoted as matrix $\mathbf{y}_{op,-p} \in \mathbb{R}^{n_{obs,p} \times (p-1)}$.

When working with y_p, $\mathbf{y}_{-p}$ are treated as covariates. A prediction model must be chosen to predict each value of y_p given the corresponding values in $\mathbf{y}_{-p}$. Wei et al. (2018) uses elastic net regression, which is a linear model that combines L1 and L2 regularization to prevent overfitting and handle multicollinearity. However, the relationships between stellar parameters may be non-linear and complex, so a more flexible model such as a random forest regressor[3] or neural network could be used instead to capture these relationships more effectively.

The chosen regressor should be trained column-by-column. For each feature y_p, the regressor is trained to predict the observed values $y_{op,p}$ given $y_{op,-p}$. If one of $\mathbf{y}_{op,-p}$ is an imputed value, it is treated as an observed value while training the regressor. This is an advantageous assumption, as it allows the regressor to be influenced by imputed values drawn from the known parameter distributions across all stellar types. This allows for some accounting of observational bias in the dataset. Iterations of the regressor will also always be primarily influenced by observed values, which prevents the process from being dominated by the initial imputed values.

After training, the regressor is used to make new predictions $y_{mp,p}^{predicted}$ for the imputed values $y_{mp,p}$ given $\mathbf{y}_{mp,-p}$.

[3] This approach is similar to MissForest imputation (Stekhoven & Bühlmann 2012) though with added stochasticity (discussed later in the algorithm) and initial values drawn from a real distribution rather than using the column mean.

Returning to the Wei et al. (2018) framework, we now have new predictions $y_{mp,p}^{predicted}$ and the old imputed values $y_{mp,p}^{old}$, both of the same length $n_{miss,p}$. The uncertainty of the predictions can be estimated by calculating the root mean squared deviation (RMSD):

$$s^2 = \frac{1}{n_{miss,p}} \sum_{n_{miss,p}} \left(y_{mp,p}^{old} - y_{mp,p}^{predicted}\right)^2 \tag{2}$$

The new imputed values $y_{mp,p}^{new}$ can then be drawn from a normal distribution centered at the predicted values with variance s^2:

$$y_{mp,p}^{new} \sim \mathcal{N}\left(y_{mp,p}^{predicted}, s^2\right) \tag{3}$$

This step introduces stochasticity into the imputation process, allowing for a more realistic representation of the uncertainty associated with the imputed values. By drawing from a distribution rather than using the predicted values directly, we can better capture the variability in the data and avoid overfitting to the observed values.

At this step, Wei et al. (2018) truncates the normal distribution for application to left-censored data. However, as we cannot generalize stellar host data to be left-censored, an untruncated normal distribution is used.

This process is repeated for each feature with missing values, and the entire procedure is iterated until convergence, which can be defined as the point at which the imputed values stabilize or a maximum number of iterations is reached. By using this modified GSimp approach, we can effectively impute missing values in the stellar host dataset while accounting for the complex relationships between parameters and the non-random nature of the missing data.

APPENDIX

A. COMPLETE CASE PCA RESULTS

A.1. *PCA with* $\tau = 0.50$

Included parameters: `sy_pnum`, `sy_snum`, `st_teff`, `st_rad`, `st_mass`, and `st_met_FeH`. Data matrix $\mathbf{X} \in \mathbb{R}^{2359\times 6}$. The raw results are as follows:

```
Principal Component 1:
  Eigenvalue: 1.4827
  Proportion of Variance: 24.7005%
  Contribution of Features:
             sy_pnum: -0.2487
             sy_snum:  0.1353
             st_teff:  0.3366
              st_rad:  0.5344
             st_mass:  0.7214
           st_met_FeH:  0.0242
Principal Component 2:
  Eigenvalue: 1.2928
```

```
  Proportion of Variance: 21.5373%
  Contribution of Features:
             sy_pnum:  0.0566
             sy_snum:  0.0334
             st_teff: -0.6073
              st_rad:  0.5143
             st_mass: -0.0643
          st_met_FeH: -0.5985
Principal Component 3:
  Eigenvalue: 1.0217
  Proportion of Variance: 17.0217%
  Contribution of Features:
             sy_pnum: -0.5170
             sy_snum:  0.8144
             st_teff: -0.1505
              st_rad: -0.1613
             st_mass: -0.1421
          st_met_FeH:  0.0260
Principal Component 4:
  Eigenvalue: 0.9501
  Proportion of Variance: 15.8287%
  Contribution of Features:
             sy_pnum: -0.8079
             sy_snum: -0.5512
             st_teff: -0.0916
              st_rad:  0.0524
             st_mass: -0.1729
          st_met_FeH:  0.0493
Principal Component 5:
  Eigenvalue: 0.8479
  Proportion of Variance: 14.1252%
  Contribution of Features:
             sy_pnum:  0.1091
             sy_snum:  0.0096
             st_teff: -0.5352
              st_rad:  0.2790
             st_mass:  0.0524
          st_met_FeH:  0.7881
Principal Component 6:
  Eigenvalue: 0.4074
  Proportion of Variance: 6.7866%
  Contribution of Features:
             sy_pnum: -0.0551
```

```
            sy_snum: -0.1161
            st_teff: -0.4477
             st_rad: -0.5859
            st_mass:  0.6501
         st_met_FeH: -0.1308
```

A.2. *PCA with* $\tau = 0.70$

Included parameters: `sy_pnum`, `sy_snum`, `st_teff`, `st_rad`, `st_mass`, `st_met_FeH`, `st_logg`, `st_age`, and `st_dens`. Data matrix $\mathbf{X} \in \mathbb{R}^{583\times 9}$. The raw results are as follows:

```
Principal Component 1:
  Eigenvalue: 4.0146
  Proportion of Variance: 44.5298%
  Contribution of Features:
            sy_pnum: -0.1036
            sy_snum: -0.0421
            st_teff:  0.4388
             st_rad:  0.4173
            st_mass:  0.4818
         st_met_FeH:  0.1299
            st_logg: -0.4587
             st_age: -0.1634
            st_dens: -0.3671
Principal Component 2:
  Eigenvalue: 1.1537
  Proportion of Variance: 12.7974%
  Contribution of Features:
            sy_pnum:  0.2363
            sy_snum:  0.4942
            st_teff:  0.1891
             st_rad: -0.0552
            st_mass:  0.0936
         st_met_FeH: -0.5313
            st_logg:  0.1009
             st_age: -0.5895
            st_dens:  0.1111
Principal Component 3:
  Eigenvalue: 1.0989
  Proportion of Variance: 12.1888%
  Contribution of Features:
            sy_pnum: -0.7382
            sy_snum:  0.5564
            st_teff: -0.0709
             st_rad: -0.0334
```

```
            st_mass:  0.0087
         st_met_FeH:  0.3287
            st_logg:  0.0594
             st_age: -0.1103
            st_dens:  0.1244
Principal Component 4:
  Eigenvalue: 0.9416
  Proportion of Variance: 10.4438%
  Contribution of Features:
            sy_pnum: -0.0742
            sy_snum:  0.4042
            st_teff: -0.0301
             st_rad:  0.2025
            st_mass: -0.0860
         st_met_FeH: -0.5162
            st_logg: -0.2252
             st_age:  0.6667
            st_dens: -0.1420
Principal Component 5:
  Eigenvalue: 0.7843
  Proportion of Variance: 8.6992%
  Contribution of Features:
            sy_pnum: -0.5663
            sy_snum: -0.5204
            st_teff:  0.0104
             st_rad:  0.1387
            st_mass: -0.0239
         st_met_FeH: -0.5281
            st_logg: -0.0412
             st_age: -0.1531
            st_dens:  0.2909
Principal Component 6:
  Eigenvalue: 0.6417
  Proportion of Variance: 7.1181%
  Contribution of Features:
            sy_pnum:  0.2415
            sy_snum:  0.0993
            st_teff: -0.3310
             st_rad:  0.5747
            st_mass: -0.0504
         st_met_FeH:  0.1665
            st_logg: -0.3135
             st_age: -0.0790
```

```
            st_dens:  0.5977
Principal Component 7:
  Eigenvalue: 0.3111
  Proportion of Variance: 3.4503%
  Contribution of Features:
            sy_pnum: -0.0622
            sy_snum: -0.0159
            st_teff: -0.5741
             st_rad:  0.2306
            st_mass: -0.3287
         st_met_FeH: -0.0789
            st_logg: -0.0805
             st_age: -0.3531
            st_dens: -0.6064
Principal Component 8:
  Eigenvalue: 0.0443
  Proportion of Variance: 0.4911%
  Contribution of Features:
            sy_pnum: -0.0038
            sy_snum:  0.0033
            st_teff: -0.0534
             st_rad: -0.5767
            st_mass: -0.1679
         st_met_FeH: -0.0050
            st_logg: -0.7863
             st_age: -0.1023
            st_dens:  0.0872
Principal Component 9:
  Eigenvalue: 0.0254
  Proportion of Variance: 0.2815%
  Contribution of Features:
            sy_pnum:  0.0069
            sy_snum: -0.0105
            st_teff: -0.5690
             st_rad: -0.2134
            st_mass:  0.7825
         st_met_FeH: -0.1165
            st_logg:  0.0210
             st_age:  0.0656
            st_dens:  0.0071
```

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